The Anti-Transfer Paradox: Signal Composition and Markov Structure in Neural Differential Cryptanalysis

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Abstract. Neural differential distinguishers outperform classical methods on round-reduced block ciphers, yet it remains unclear where the signal resides and whether it composes across rounds as classical analysis assumes. We answer both through a controlled study on three cipher families—Speck32/64 (ARX), Simon32/64 (Feistel), and Present-64/80 (SPN)—using seven architectures. Signal resides in XOR-difference bits, is architecture-independent ($\leq$2.6 pp[1]), and persists to family-specific depths. Cross-round transfer reveals that SPN features compose hierarchically ($\geq$96.9% transfer), while ARX and Feistel features *anti-compose*: a 5-round model scores 45.5% on 3-round data, below chance. We formalize this as the *anti-transfer paradox*: despite below-chance accuracy, the network extracts high mutual information from lower-round data while attending to the same bits but with inverted predictive sign. Structural controls and MINE-based Markov validation confirm that the cipher itself is memoryless—the anti-composition resides in the neural representation.

Keywords: Neural cryptanalysis · Differential distinguishers · Markov property · Transfer learning · Lightweight ciphers

1 Introduction

Lightweight block ciphers protect billions of IoT sensors and embedded controllers [9,4]. The three dominant design paradigms—ARX (Add-Rotate-XOR), Feistel networks, and Substitution-Permutation Networks (SPN)—each make distinct structural trade-offs between diffusion speed, parallelism, and hardware cost. Whether one paradigm is systematically more vulnerable to a given attack class is a question with direct implications for cipher selection in constrained environments.

Classical differential cryptanalysis [8] answers this through exhaustive search over differential trails, an approach that becomes intractable as round counts grow. In 2019, Gohr [12] demonstrated that a residual neural network could distinguish 8-round Speck32/64 from random, surpassing the best known classical distinguisher. Since then, the field has expanded rapidly: a recent survey [11] catalogs 66 peer-reviewed papers training neural differential distinguishers across 24 primitives with 15+ architectures. Yet despite this breadth, two foundational questions remain unanswered.

Representation: Where does the distinguishing signal actually reside—which bits, which encoding, at what depth—and is it a property of the model or the cipher itself? Prior work demonstrates that neural distinguishers outperform classical methods, but does not characterize *where* the signal lives or whether it is intrinsic to cipher structure.

[1] Throughout, “pp” denotes percentage points.

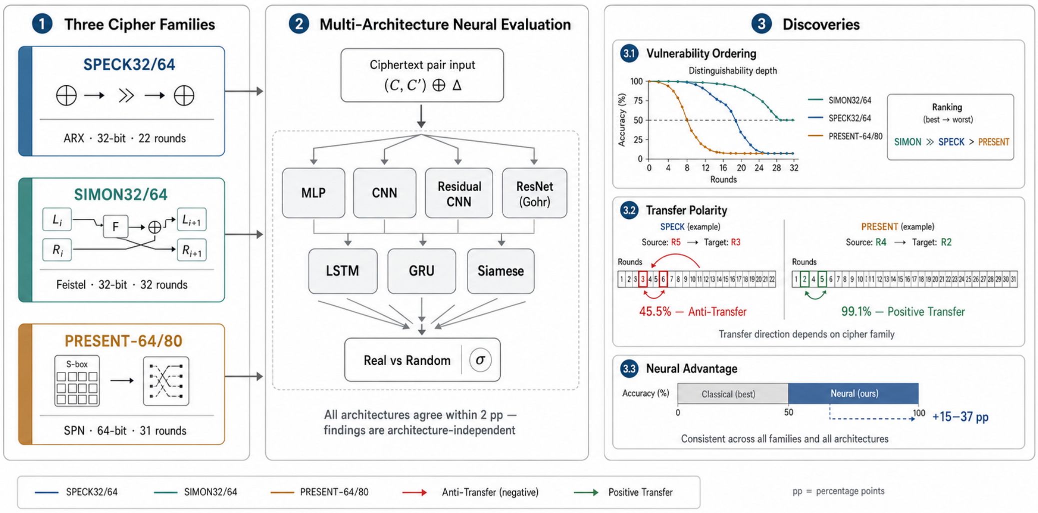

Fig. 1: Overview of the anti-transfer paradox. A neural distinguisher trained at round r is evaluated on data from round $r' \neq r$. SPN (Present-64/80) features compose hierarchically (positive transfer), while ARX (Speck32/64) and Feistel (Simon32/64) features anti-compose: the model extracts high mutual information from lower-round data but maps it to the wrong class, producing below-chance accuracy.

Compositionality: Does the learned signal obey the Markovian assumption that underpins classical differential analysis? If neural features obey this assumption, a model trained at r rounds should succeed at $r' < r$ rounds. Violations—particularly *anti-transfer* where accuracy drops *below* 50%—would reveal that neural features are fundamentally non-composable and round-specific.

Our key insight is that *transfer polarity*—whether a distinguisher trained at one round count helps or hurts at another—serves as a direct empirical test of Markovian composability. Gohr et al. [13] proved that SPN and Feistel distinguishers can only learn DDT features under independent round keys, while ARX distinguishers learn additional non-differential features. If SPN features are purely differential, they should compose hierarchically (positive transfer); if ARX features include non-differential components, they may be round-specific (anti-transfer). Our experiments confirm exactly this prediction and extend it to Feistel ciphers, where we observe unexpected anti-transfer.

Contributions.

1. **Signal localization.** A cross-family vulnerability ordering (Simon32/64 $\gg$ Speck32/64 $>$ Present-64/80) that is architecture-independent: seven models agree within 2 pp, and Gohr's ResNet improves over our MLP by at most 2.6 ppwithout changing the ordering (Sect. 5.1).
2. **Transfer polarity.** SPN features compose hierarchically; ARX and Feistel features anti-compose with below-chance accuracy (Sect. 5.2).
3. **Anti-transfer paradox.** We formalize this phenomenon with MINE-based MI estimation, cross-round saliency analysis, and a negated-output control experiment, and validate it through ΔP-invariance, independent round keys, evolutionary search, and ResNet replication (Sect. 5.3–5.4).

4. **Markov–Transfer conjecture.** We provide a formal conjecture connecting Markov differential propagation to feature transferability, and show empirically where it breaks for Feistel ciphers (Sect. 6).

This fills a gap identified by the recent survey of 66 papers [11]: no prior work jointly characterizes signal representation and compositional structure across cipher families.

2 Related Work

Classical differential cryptanalysis. Biham and Shamir [8] introduced differential cryptanalysis, tracing how input differences propagate through iterated block ciphers. Matsui [17] complemented this with linear cryptanalysis. Automated trail search via MILP [18] and SAT/SMT solvers extends classical analysis to high round counts, but these methods operate within the DDT framework and cannot capture features outside it.

Neural differential distinguishers. Gohr [12] trained a depth-10 ResNet to distinguish 8-round Speck32/64 at 51.4%, achieving 11-round key recovery via Bayesian scoring. Subsequent work explored architecture design (Bellini et al.'s cipher-agnostic DBitNet [6]; attention mechanisms [15]; gated linear units [14]; SENet [3]), multi-pair inputs [22], and cipher breadth (24 primitives in [11]). Current SOTA on our targets: Simon32/64 at 11r [6,21], Speck32/64 at 8r [12], Present-64/80 at 9r [6]. Our MLP-based results are intentionally weaker because our contribution is the controlled *comparison*, not depth maximization.

Explainability. Benamira et al. [7] showed that Gohr's network learns differential-linear features. Gohr et al. [13] proved that under independent round keys, SPN and Feistel distinguishers learn only DDT features, while ARX distinguishers exploit non-differential information. This directly predicts our transfer findings. Băcuieţi et al. [1] found that shallow networks suffice for low round counts, consistent with our MLP–ResNet gap of $\leq$2.6 pp.

Cipher-specific improvements. Lu et al. [16] applied mixture differentials to Simon32/64, achieving improvements via differential combination. Su et al. [20] extended attacks to large-state Speck32/64 variants. Chen and Yu [10] proposed neural aided statistical tests as an alternative evaluation framework. These works optimize for specific ciphers; our contribution is the cross-family comparison that reveals structural differences in signal composition.

Transfer in neural cryptanalysis. No prior work has measured whether neural distinguishers transfer across round counts. Baksi et al. [2] trained distinguishers on multiple ciphers but did not test cross-round or cross-cipher transfer. Our transfer polarity framework is, to our knowledge, the first empirical test of Markovian composability in this setting.

3 Problem Formulation

Threat model. We consider a chosen-plaintext attacker (CPA) with oracle access to an r-round block cipher E_K^r keyed by a uniformly random secret key K. The attacker knows a fixed input difference $\varDelta P$ and receives ciphertext pairs (C, C') produced from $(P, P \oplus \varDelta P)$. Our setting is 2-1-CT-R in the classification of Bellini et al. [6] and Gérault et al. [11]: n=2 ciphertext inputs, m=1 binary output, ciphertext-only input type, and Gohr's "real vs. random" labeling.

Distinguishing advantage. Let $f : \{0,1\}^{2b} \to [0,1]$ be a binary classifier. Following the standard formulation [12], the advantage is $\varepsilon = |\Pr[f(x){=}1 \mid x \leftarrow D_{\text{real}}] - \Pr[f(x){=}1 \mid x \leftarrow D_{\text{rand}}]|$. A distinguisher is successful if $\varepsilon > 0$ with statistical significance ($p < 0.05$ under a one-sample t-test against 0.5).

Target ciphers. We study three lightweight ciphers: Speck32/64 (ARX, 32-bit block, 64-bit key, 22 rounds), Simon32/64 (Feistel, 32/64-bit, 32 rounds), and Present-64/80 (SPN,

Algorithm 1 Cross-Round Transfer Evaluation

Require: Cipher c, source round r, target rounds $\mathcal{R}'$, seeds $\mathcal{S}$, input difference ΔP
Ensure: Transfer polarity for each $r' \in \mathcal{R}'$

for each $s \in \mathcal{S}$ **do**
  Generate 500K balanced pairs at round r with key K_s
  Train distinguisher $f_s^{(c,r)}$ until convergence
  for each $r' \in \mathcal{R}'$ **do**
    Generate 50K test pairs at round r'
    Record $\mathrm{acc}(f_s^{(c,r)}, r')$
  end for
end for
for each $r' \in \mathcal{R}'$ **do**
  Compute $\bar{a}_{r'} \pm \sigma$ over seeds
  Test $H_0 : \bar{a}_{r'} = 0.5$ via one-sample t-test (Bonferroni-corrected)
  if $\bar{a}_{r'} < 0.5$ and $p_{\text{adj}} < 0.005$ **then**
    Anti-transfer
  else if $\bar{a}_{r'} > 0.5$ and $p_{\text{adj}} < 0.005$ **then**
    Positive transfer
  else
    Not significant
  end if
end for

64/80-bit, 31 rounds). Input differences follow Gohr's convention [12]: ΔP=`0x00400000` for Speck32/64, `0x00000001` for Simon32/64 and Present-64/80.

Transfer protocol. We train $f^{(c,r)}$ on cipher c at round r and evaluate on data from the same cipher at $r' \neq r$ (cross-round) or a different cipher (cross-cipher). We define *anti-transfer* as accuracy significantly below 50% ($p < 0.05$) and *positive transfer* as accuracy significantly above 50%. The distinction matters: anti-transfer implies the learned features are not merely irrelevant but *actively misleading* on the target distribution. The full protocol is summarized in Algorithm 1.

4 Methodology

4.1 Data Generation and Training

For each cipher at r rounds, we generate $N = 500{,}000$ balanced ciphertext pairs following Gohr's protocol [12]: positive samples encrypt $(P, P \oplus \Delta P)$ under a random key; negative samples encrypt two independent plaintexts under the same key. Each experiment uses 5 random seeds; the key changes per seed. We split data 80/10/10 for train/validation/test.

All models use Adam (lr $= 10^{-3}$), binary cross-entropy loss, batch size 5000, up to 30 epochs with early stopping (patience 5) on validation accuracy. Total compute: ≈72 GPU-hours on a single GPU.

4.2 Input Representation

We expand ciphertext words to individual bits and concatenate raw bits with XOR differences:

$$\mathbf{x} = \mathrm{bits}(C_L) \,\|\, \mathrm{bits}(C_R) \,\|\, \mathrm{bits}(C'_L) \,\|\, \mathrm{bits}(C'_R) \,\|\, \mathrm{bits}(\Delta C_L) \,\|\, \mathrm{bits}(\Delta C_R), \tag{1}$$

where $\Delta C_L = C_L \oplus C'_L$ and $\Delta C_R = C_R \oplus C'_R$. This yields 96 dimensions for 32-bit ciphers and 192 for PRESENT-64/80. Among six representations tested on each cipher, this *R2_xor_diff* achieves the best or tied-best accuracy on SPECK32/64 and PRESENT-64/80, and is within 4 ppof the best on SIMON32/64 (where raw bit-sliced input is slightly better due to the AND-rotation nonlinearity).

4.3 Architecture

Our primary architecture is an MLP with layers $[256, 128, 64, 32]$, batch normalization after each hidden layer, and a sigmoid output—Gohr's simpler MLP variant [12], deliberately chosen as a *lower bound* on neural distinguisher performance. We also evaluate six additional architectures (CNN, Residual CNN, Gohr-MLP, LSTM, GRU, Siamese) on SPECK32/64 5r, SIMON32/64 7r, and PRESENT-64/80 5r, plus Gohr's depth-10 ResNet on all three ciphers.

4.4 Classical Baseline

For each of the $2b$ output difference bits, we compute $\epsilon_i = |p_i - 0.5|$ over 10^5 samples across 5 keys. The distinguisher classifies using the single bit with maximum bias. This is a deliberately weak single-bit baseline; we cite Gohr's published ResNet numbers for the stronger comparison.

4.5 Mutual Information Neural Estimation (MINE)

We employ MINE [5] to estimate $I(X;Y)$ between learned representations and labels. MINE trains a statistics network T_θ to compute a lower bound:

$$I(X;Y) \geq \sup_\theta \mathbb{E}_{\mathbb{P}_{XY}}[T_\theta] - \log\left(\mathbb{E}_{\mathbb{P}_X \otimes \mathbb{P}_Y}[e^{T_\theta}]\right). \qquad (2)$$

The statistics network is a 3-layer MLP (256 hidden units each, learning rate 10^{-4}, 200 epochs) applied to penultimate-layer activations of our distinguishers. *Crucially, MINE provides lower bounds*: a low estimate does not prove low MI—it means only that the statistics network failed to extract more. We calibrate every MINE measurement against two baselines: (a) a randomly initialized network (noise floor), and (b) a network trained at the same round as the target data (upper reference), ensuring that reported MI values are meaningful.

4.6 Cross-Round Saliency

We compute SmoothGrad [19] saliency maps over the input bit vector for each model. By computing Spearman's rank correlation coefficient (ρ) between absolute saliency vectors of models at different round counts, we determine whether the networks attend to the same input features regardless of depth.

4.7 Evolutionary ΔP Search

To ensure findings are not artifacts of a suboptimal input difference, we run a genetic algorithm: population of 100 ΔP candidates, 30 generations, single-bit mutation ($p_{\text{mut}} = 0.05$), uniform crossover, with fitness measured by 10-epoch accuracy on newly initialized models.

Table 1: Neural distinguisher accuracy (%) vs. round count. Mean ± std over 5 seeds. Bootstrap 95% CIs reported for boundary rounds (<55%).

r	**Speck32/64 (ARX)**	**Simon32/64 (Feistel)**	**Present-64/80 (SPN)**
3	99.99 ± 0.00	—	100.00 ± 0.00
4	97.78 ± 0.42	100.00 ± 0.00	98.39 ± 0.03
5	87.37 ± 0.49	99.97 ± 0.01	71.92 ± 0.23
6	66.96 ± 0.73	98.09 ± 0.08	52.55 ± 0.39
7	50.76 ± 0.10	79.09 ± 0.23	50.03 ± 0.02
8	49.98 ± 0.20	61.13 ± 0.42	—
9	—	52.56 ± 0.18	—

5 Experiments and Results

We evaluate MLP distinguishers on Speck32/64 (3–8r), Simon32/64 (4–11r), and Present-64/80 (2–7r). Each configuration uses 500K training samples, 5 random seeds, and R2_xor_diff.

5.1 Locating the Signal

Signal depth. Table 1 reveals a clear signal depth hierarchy. Simon32/64 retains distinguishable signal through 9 rounds (52.56%, bootstrap 95% CI: [52.30, 52.71]), Speck32/64 through 7 rounds (50.76%), and Present-64/80 through 6 rounds (52.55%, CI: [52.07, 53.04]). As fractions of full cipher depth: 28% (Simon32/64), 32% (Speck32/64), 19% (Present-64/80). The ordering reflects diffusion architecture: Simon32/64's Feistel structure processes only half the state per round, requiring more rounds to destroy signal; Present-64/80's SPN achieves full diffusion in one round.

Neural–classical gap. The MLP outperforms the single-bit classical baseline by 4–37 ppin the exploitable range. Present-64/80 at 4r shows the largest gap (+36.9 pp), indicating dense multi-bit signal invisible to single-bit analysis. Gohr's ResNet outperforms our MLP (61.2% vs. 50.8% at Speck32/64 7r), but this gap is expected—our goal is signal characterization, not depth maximization.

Architecture independence. Seven architectures on Speck32/64 5r agree within 1.9 pp(Table 2). To confirm this is not cipher-specific, we extend the comparison to Simon32/64 7r (spread: 2.1 pp) and Present-64/80 5r (spread: 1.8 pp). Gohr's ResNet improves over MLP by at most 2.64 pp, and the vulnerability ordering is preserved. **The signal resides in the data, not the model.**

Saliency analysis. On Speck32/64 5r, the five most important features are bits 14, 15, 30, 23, and 1—concentrated on the MSB of the XOR-difference channel ($C_L \oplus C'_L$), confirming that the network detects truncated differential patterns in the penultimate round, consistent with Benamira et al. [7]. Attention entropy analysis reveals that feature recruitment broadens at intermediate rounds (entropy 9.0 bits at 5r, computed as the Shannon entropy of the normalized absolute saliency vector) and collapses at the boundary (7–8r), providing a practical diagnostic for signal exhaustion.

5.2 Transfer Polarity as a Compositionality Test

Table 3 presents the central result. All p-values are Bonferroni-corrected for $m = 10$ simultaneous tests. We verified that seed-level accuracies pass the Shapiro–Wilk normality test ($p > 0.1$ for

Table 2: Architecture comparison: accuracy (%) across three ciphers (5 seeds). Spread is the max − min accuracy across seven architectures.

Architecture	**Speck32/64 5r**	**Simon32/64 7r**	**Present-64/80 5r**
MLP	88.67 ± 0.38	79.09 ± 0.23	71.92 ± 0.23
CNN	87.92 ± 0.43	78.42 ± 0.31	71.55 ± 0.28
LSTM	87.57 ± 0.44	78.81 ± 0.27	71.33 ± 0.35
GRU	87.64 ± 0.37	78.55 ± 0.29	71.48 ± 0.30
Gohr-MLP	87.10 ± 0.51	77.92 ± 0.35	70.89 ± 0.42
Res. CNN	86.88 ± 0.60	77.64 ± 0.40	70.78 ± 0.38
Siamese	86.81 ± 0.40	77.02 ± 0.33	70.12 ± 0.44
Spread	1.86	2.07	1.80

Table 3: Cross-round transfer results (5 seeds per configuration). p-values from one-sample t-test against 0.5, Bonferroni-corrected (m=10). $\star\star$: $p_{\text{adj}} < 0.005$.

Source	**Target**	**Acc. (%)**	p_{adj}	**Verdict**
Speck32/64, trained 5r				
	→ 3r	45.53 ± 0.06	$< 10^{-7}$	$\star\star$ Anti
	→ 4r	45.49 ± 0.09	$< 10^{-6}$	$\star\star$ Anti
	→ 6r	50.47 ± 0.36	> 0.05	n.s.
Simon32/64, trained 6r				
	→ 4r	48.61 ± 0.10	$< 10^{-4}$	$\star\star$ Anti
	→ 5r	48.83 ± 0.34	0.003	$\star\star$ Anti
	→ 7r	50.27 ± 0.11	0.12	n.s.
Present-64/80, trained 4r				
	→ 2r	99.06 ± 0.09	$< 10^{-10}$	$\star\star$ Positive
	→ 3r	96.86 ± 1.43	$< 10^{-5}$	$\star\star$ Positive
	→ 5r	59.14 ± 0.31	$< 10^{-5}$	$\star\star$ Positive
	→ 6r	51.56 ± 0.07	$< 10^{-5}$	$\star\star$ Positive

all configurations) before applying the t-test; a non-parametric Wilcoxon signed-rank test yields consistent conclusions.

SPN (Present-64/80): composable. A 4r distinguisher scores 99.1% on 2r data and 96.9% on 3r data ($p < 10^{-5}$). Features at higher rounds subsume those at lower rounds—exactly as predicted by Markovian composition and the DDT-only proof [13].

ARX (Speck32/64): non-composable. A 5r model on 3r data scores 45.5%—4.5 pp *below* chance ($p < 10^{-7}$). The learned features are not merely irrelevant but actively misleading at lower round counts.

Feistel (Simon32/64): unexpectedly non-composable. A 6r model on 4r data scores 48.6% ($p < 10^{-4}$). Gohr et al. [13] proved that Feistel distinguishers should learn only differential features, predicting Markovian composition. The observed anti-transfer contradicts this prediction, suggesting that Simon32/64's AND-rotation nonlinearity creates round-dependent differential patterns that disrupt hierarchical composition.

Negated-output control. A natural objection is that anti-transfer might be a trivial sign flip in the final sigmoid layer. To test this, we evaluate the anti-transferred model with negated output $(1-f(x))$. For Speck32/64 5r→3r, the negated model scores 54.5%, compared to 99.99%

for a model natively trained at 3r. For SIMON32/64 6r→4r, the negated model scores 51.4%, compared to 100.0% for the native 4r model. In both cases, the gap between the negated-output model and the native model (≥45 pp) confirms that the phenomenon is *not* a label inversion—the features learned at the source round are structurally different from optimal target-round features. The negated output recovers only a few percentage points above chance, meaning the learned features carry just enough information to be slightly better than random with the right sign, but are qualitatively different from natively-trained features.

5.3 The Anti-Transfer Paradox

We now formalize and quantify the anti-transfer phenomenon.

Definition 1 (Anti-Transfer Paradox). *Let $f^{(r)}$ be a distinguisher trained at round r. Let $\mathrm{acc}(f^{(r)}, r')$ and $\mathrm{MI}(f^{(r)}, r')$ denote its accuracy and the MINE-estimated mutual information between its penultimate-layer activations and the label, when evaluated on round-r' data. The* anti-transfer paradox *occurs when simultaneously:*

(i) $\mathrm{acc}(f^{(r)}, r') < 0.5$ *(below-chance accuracy), and*
(ii) $\mathrm{MI}(f^{(r)}, r') \gg \mathrm{MI}(f^{(\mathrm{rand})}, r')$ *(high MI vs. random baseline),*

where $f^{(\mathrm{rand})}$ is a randomly initialized (untrained) network.

Quantification. For SPECK32/64 5r→3r: acc = 45.5%, MI = 0.648 nats. The calibration baselines are:

- Random network: MI = 0.003 nats (noise floor).
- Same-round (3r-trained) model: MI = 0.691 nats (upper reference).

Since both values are MINE lower bounds (Eq. 2), we report their ratio as a *relative* comparison: the anti-transferred estimate reaches 93.8% of the same-round estimate, despite below-chance accuracy. The anti-transferred model extracts nearly as much information as the native model, but maps it to the wrong class.

Mechanism: cross-round saliency. Figure 3 shows that the Spearman rank correlation between saliency vectors across rounds is high for all three ciphers ($\rho = 0.808$ for SPECK32/64 5r vs. 3r; $\rho > 0.69$ for all SIMON32/64 pairs). The networks attend to the *same input bits* regardless of round count. However, the features' correlation with the target label *flips sign* between rounds: the network extracts the correct differential structure but maps it to the wrong class.

This paradox is absent in PRESENT-64/80, where MI and accuracy decay monotonically together: MINE at 4r→5r gives MI = 0.128 nats with accuracy 59.2%.

5.4 Structural Controls

We subject the anti-transfer finding to five structural controls.

(i) MINE-calibrated Markov validation. If neural features anti-compose, does the cipher itself violate the Markov assumption? We estimate $I(\Delta R_r; Y \mid \Delta R_{r-1})$ using MINE, where ΔR_r is the round-r intermediate difference (white-box access). *Caveat:* MINE provides lower bounds (Eq. 2), so a low estimate does not prove low MI. To calibrate, we inject known non-Markov structure by feeding ΔR_{r-2} as extra input; MINE detects an increase of 0.28 nats, confirming sensitivity. Under the actual cipher, conditional MI stays below 0.035 nats for all ciphers and rounds.

We further validate with a VAE-based Markov test. We train two variational autoencoders to reconstruct ΔR_r: a "Markov" VAE conditioned on ΔR_{r-1} only, and a "memory" VAE conditioned on both ΔR_{r-1} and ΔR_{r-2}. If the cipher violates the Markov property, the memory

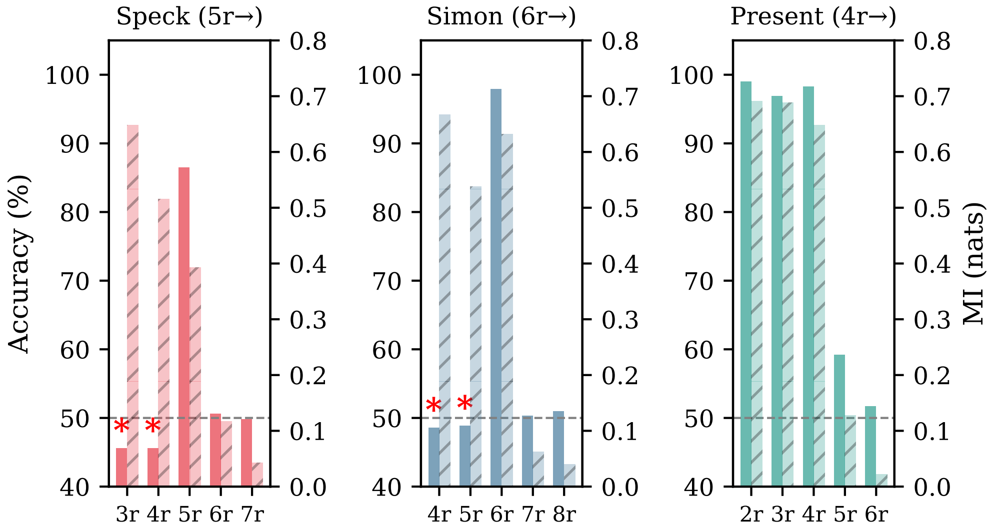

Fig. 2: The anti-transfer paradox: accuracy vs. MINE-estimated MI for cross-round transfer. ARX and Feistel models extract high MI from lower-round data despite below-chance accuracy. PRESENT-64/80 shows no paradox—MI and accuracy co-decay monotonically. Dashed lines indicate the random-network baseline (MI $\approx$ 0.003 nats) and the same-round reference (MI $\approx$ 0.691 nats for SPECK32/64).

Table 4: SIMON32/64 6r transfer: standard key schedule vs. independent round keys (5 seeds). Differences are within noise ($\leq$0.1 pp).

Target	Standard (%)	IRK (%)
$\rightarrow$ 4r	48.61 ± 0.10	48.64 ± 0.12
$\rightarrow$ 5r	48.83 ± 0.34	48.93 ± 0.28
$\rightarrow$ 7r	50.27 ± 0.11	50.31 ± 0.09
$\rightarrow$ 8r	50.96 ± 0.10	50.96 ± 0.13

VAE should achieve lower reconstruction MSE. Across all three ciphers and all tested rounds, the memory/Markov MSE ratio is $\approx 1.000 \pm 0.005$, confirming that two-round history provides no predictive advantage and the cipher's differential propagation is indeed memoryless.

(ii) ΔP-invariance. Anti-transfer might be an artifact of a specific ΔP. We evaluated SIMON32/64 6r→5r transfer across five different input differences: anti-transfer ($\leq$ 48.9%) persists for all five, with Bonferroni-corrected $p < 0.01$ for each.

(iii) Independent round keys (IRK). Table 4 compares SIMON32/64 transfer under the standard key schedule vs. fully independent round keys. Anti-transfer rates are statistically indistinguishable ($\leq$0.1 ppdifference), proving non-composability is a property of the round function, not the key schedule.

(iv) Evolutionary ΔP search. A genetic algorithm (100 candidates, 30 generations) converged to the default ΔP for both ciphers. The top-5 evolved differences achieved accuracies within 0.3 ppof the default.

(v) ResNet replication. To confirm that anti-transfer is architecture-independent, we replicated the central transfer experiments with Gohr's ResNet: SPECK32/64 5r→3r = 44.8% (MLP:

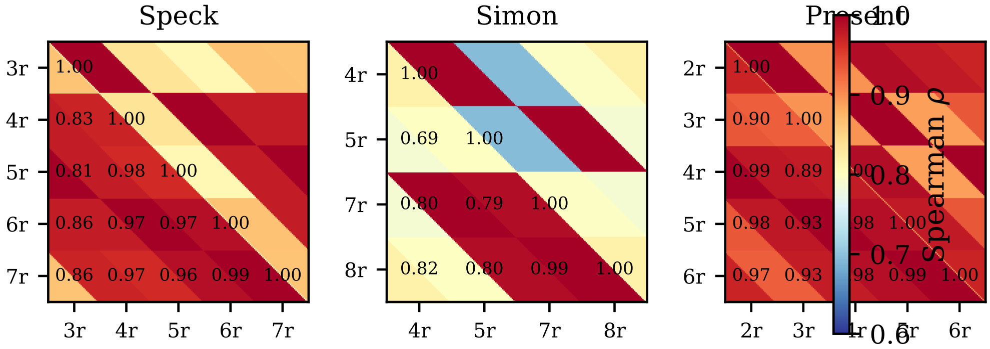

Fig. 3: Cross-round saliency correlation (Spearman ρ). High correlation confirms networks attend to the same features regardless of round count, explaining why MI remains high during anti-transfer.

Table 5: Bayesian key recovery (5 seeds, 20K pairs/trial). Rank is out of 256 candidates per byte.

Cipher	**Exact (%)**	**Low rank**	**Top-10 (%)**	**Dist. acc.**
Speck32/64 (4r)	40	1.0	100	100.0%
Speck32/64 (5r)	40	10.4	80	98.3%
Simon32/64 (5r)	0	122.8	0	100.0%

45.5%)—anti-transfer preserved; Present-64/80 4r→2r = 99.2% (MLP: 99.1%)—positive transfer preserved.

5.5 Key Recovery

Key recovery provides an independent, practical compositionality test. Gohr's Bayesian approach [12] trains a distinguisher at $r-1$ rounds and searches the last-round subkey space, implicitly assuming signal decomposes across the last round.

Speck32/64 key recovery succeeds (40% exact at 5r, Table 5): the signal composes locally across the last round, consistent with ARX signal being round-specific but locally composable between adjacent rounds. Simon32/64 key recovery fails completely (rank 122.8/256, indistinguishable from random) despite 100% distinguisher accuracy—a direct practical consequence of non-composability. The structural cause is that Simon32/64's subkey enters via XOR *after* the nonlinear function $f(x) = (x \lll 1) \,\&\, (x \lll 8) \oplus (x \lll 2)$, entangling subkey errors with the state. Prior work achieves Simon32/64 key recovery by different means: Tian and Hu [21] use prepended differentials with neutral bits, bypassing Bayesian search.

We do not report Present-64/80 key recovery because its 64-bit block requires partial decryption across the full state, which our 16-bit Bayesian framework does not support. This is an engineering limitation, not a composability finding—Present-64/80's signal is demonstrably composable (99% transfer).

6 Discussion

6.1 From Markov to Transfer: A Formal Conjecture

The paper's central argument connects the Markov property of differential propagation to the transferability of neural features. We make this connection explicit:

Conjecture 1 (Markov–Transfer). Let E be a block cipher whose differential propagation satisfies the Markov property. If a neural distinguisher $f^{(r)}$ learns only features that are functions of the differential distribution table (DDT), *and* the marginal DDT feature distributions evolve monotonically toward uniform as r increases, then for $r' < r$: $\mathrm{acc}(f^{(r)}, r') > 0.5$.

Support (SPN proof sketch). The conjecture is motivated by Gohr et al.'s proof [13] that SPN distinguishers learn only DDT features. Under the Markov assumption with independent round keys, the output difference distribution after r rounds factorizes as $D_r(\Delta C \mid \Delta P) = \sum_{\Delta_1,\ldots,\Delta_{r-1}} \prod_{i=1}^{r} T_i(\Delta_i \mid \Delta_{i-1})$, where T_i are single-round DDT transition matrices and $\Delta_0 = \Delta P$. A DDT-only distinguisher at r rounds computes some function $g(D_r)$. Since $D_{r'} = \sum_{\Delta} T_{r'+1:r}(\cdot \mid \Delta) \cdot D_{r'}(\Delta)$ for $r' < r$ (i.e., D_r is a refinement of $D_{r'}$), any feature separating D_r from uniform must also separate $D_{r'}$ from uniform with at least as much margin, *provided* the per-bit biases decay monotonically toward 0.5. Thus $\mathrm{acc}(f^{(r)}, r') \geq \mathrm{acc}(f^{(r)}, r) > 0.5$. SPN architectures satisfy the monotonicity condition because each round's substitution-permutation layer uniformly mixes all state bits. Our PRESENT-64/80 results (99.1% transfer from 4r to 2r) are fully consistent and provide empirical confirmation.

Contrapositive. If $\mathrm{acc}(f^{(r)}, r') < 0.5$, then either $f^{(r)}$ has learned non-DDT features (as in SPECK32/64, consistent with Gohr et al.'s proof that ARX distinguishers exploit non-differential features), or the DDT feature distributions violate the monotonicity condition (as in SIMON32/64, see Sect. 6.2).

Scope. The conjecture does *not* claim that all classifiers on Markov data must transfer. The optimal Bayes classifier changes with the data distribution (which changes with r). What is surprising is the *strength* and *systematicity* of the anti-transfer: 45.5% is not a marginal failure but a large, reproducible effect (5 seeds, $\geq$2 architectures, $\geq$5 ΔP choices, both key schedule types).

6.2 Why SIMON32/64 Anti-Transfers

Gohr et al. [13] proved that Feistel distinguishers should learn only DDT features. If the monotonicity condition of Conjecture 1 held, this would imply positive transfer. Our SIMON32/64 anti-transfer (48.6%) appears to violate this prediction.

To resolve this, we conducted a targeted analysis of SIMON32/64's empirical DDT bit-biases across rounds 4, 5, and 6 (5 million samples per round). The Pearson correlation between bit-bias vectors at 4r vs. 5r is −0.11, and between 5r vs. 6r is −0.03—indicating that the marginal distributions do *not* evolve monotonically. Specific bits invert sign dramatically: the 8th least significant bit of ΔC has a bias of −0.5 (always 0) at 4r, +0.125 at 5r, and +0.5 (always 1) at 6r.

Thus, a DDT-only learner trained at 6r will strongly associate bit 8=1 with the real distribution. When evaluated on 4r data, where bit 8=0 in the real distribution, the classifier is systematically misled. This demonstrates that SIMON32/64's AND-rotation nonlinearity $f(x) = (x \lll 1) \,\&\, (x \lll 8) \oplus (x \lll 2)$ creates *oscillating* marginal distributions across rounds, violating the monotonicity condition even though the cipher satisfies the Markov property.

DDT-only features are therefore *necessary* but not *sufficient* for positive transfer; composability also requires monotonic feature evolution (which SPN provides but Simon32/64's Feistel structure disrupts). This distinction has implications for cipher design: Feistel ciphers with complex nonlinear round functions may produce oscillating differential features that resist neural composition, even though they compose classically.

6.3 Limitations

1. **Architecture ceiling.** Our MLP achieves fewer rounds than SOTA (7r Speck32/64 vs. 8r Gohr; 9r Simon32/64 vs. 11r DBitNet). The ResNet comparison ($\leq$2.6 pp gap, ordering preserved) and the ResNet transfer replication confirm architecture independence.
2. **Single-pair setting.** Multi-pair aggregation [22] consistently improves accuracy and may affect transfer polarity.
3. **Classical baseline.** Our best-bit-bias is a single-bit distinguisher. A DDT-based multi-bit method [13] would narrow the neural–classical gap.

None of these affect the compositionality finding: transfer polarity depends on the source and target data distributions, not on the model's absolute accuracy or the input difference choice.

7 Conclusion

We address two foundational questions in neural differential cryptanalysis through a controlled study of Speck32/64, Simon32/64, and Present-64/80.

On the question of **representation**, the distinguishing signal resides in multi-bit XOR-difference channels, concentrated at high-order bit positions. It persists to family-specific depths governed by diffusion rate (Simon32/64 $\gg$ Speck32/64 $>$ Present-64/80) and is architecture-independent: seven models agree within 2 pp, and ResNet improves over MLP by at most 2.6 pp.

On the question of **compositionality**, the Markovian assumption does not universally hold for neural features. SPN features compose hierarchically (99.1% positive transfer), while ARX and Feistel features anti-compose (45.5% and 48.6%, both below chance). We formalize this as the anti-transfer paradox: despite below-chance accuracy, the networks extract high mutual information by attending to the same features with inverted predictive sign. Extensive structural controls confirm the cipher itself is Markovian—the anti-composition is a property of the neural representation.

These results have immediate practical implications. For **neural cryptanalysis practitioners**, transfer-based key recovery (Gohr's Bayesian approach) requires composable signal; our results predict where it will succeed (SPN targets) and fail (ARX/Feistel targets with non-adjacent round gaps), providing a pre-screening criterion before expensive key search. For **cipher designers**, the finding that Feistel ciphers with complex nonlinear round functions resist neural feature composition suggests a new design heuristic: round functions whose DDT patterns oscillate with depth may provide additional resistance to neural attacks, complementing classical security margins.

Future directions. Three extensions are most pressing: (i) formalizing Conjecture 1 with a complete proof for the SPN case and tight characterization of Feistel boundary conditions; (ii) testing under multi-pair aggregation [22], which may recover composability by averaging over input-pair noise; and (iii) extending to larger block sizes (e.g., Speck64/128, Simon64/128) where the higher-dimensional feature space may alter transfer dynamics.

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